Lab 8: The Central Dogma — DNA to Protein

BIOL-1

Name: ______________________ **Date:** ______________________

Overview

The "Central Dogma" of biology describes how genetic information flows from DNA to RNA to Protein. In this lab, you will model these processes using paper or bead simulations. This hands-on approach helps visualize the molecular machinery that is too small to see.

Learning Objectives

By the end of this lab, you will be able to:

1. Model DNA replication to create two identical daughter strands.
2. Transcribe a DNA sequence into a complementary mRNA sequence.
3. Translate an mRNA sequence into a polypeptide chain using the genetic code.
4. Predict the consequences of mutations (substitution, insertion, deletion) on the final protein structure.

Materials

- DNA Sequence Strips (provided)
- mRNA Nucleotide Cards (A, U, C, G)
- tRNA/Amino Acid Cards
- Genetic Code Chart (Codon Table)
- Tape/Scissors

Activity 1: Transcription (DNA → mRNA)

Procedure:

1. Obtain a **Template DNA Strand**. Write down its sequence below.

2. Act as **RNA Polymerase**: Match complementary RNA nucleotides to the DNA template.
    - Remember: A pairs with ***, T pairs with***, C pairs with ***, G pairs with***.
3. Write your resulting mRNA sequence. This is the **transcript**.

Data:

- **Template DNA:**
- **mRNA Transcript:**

Activity 2: Translation (mRNA → Protein)

Procedure:

1. Act as the **Ribosome**: Read your mRNA transcript in groups of three nucleotides (**codons**).
2. Use the **Genetic Code Chart** to determine which amino acid corresponds to each codon.
3. Find the matching tRNA/amino acid card and link them together to form a polypeptide chain.

Data:

- **Polypeptide Sequence (Amino Acids):**

1. Why is the order of amino acids important for the protein's function?

Activity 3: Mutation Analysis

Procedure:

1. Take your original DNA sequence.
2. **Mutate it**: Change the 4th nucleotide to a different base (a **Substitution**).
3. Transcribe and translate this new mutated sequence.

Data:

- **Mutated DNA:**
- **Mutated mRNA:**
- **Mutated Protein:**

2. Did the amino acid sequence change? If so, how?

3. What happens if you insert an extra nucleotide (Insertion) near the beginning? How does this affect the "reading frame"?

Conclusion

The flow of information relies on precise base pairing. Mutations can alter this flow, potentially changing the protein product and the organism's traits.